Named Entity Recognition in Chinese Electronic Medical Records Based on Global Pointer and Medical BERT

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Abstract

Chinese electronic medical record named entity recognition suffers from low accuracy and high complexity. In this paper, we propose a MBBLEGP model based on medical pre-training, which firstly masks the medical entities in the input medical record text data by MC-BERT to obtain the medical text word vector representations; then the word vectors are inputted into BiLSTM to extract the forward and backward local feature information to obtain the feature vectors; finally, a self-attention layer for GlobalPointer is used to Finally, rotational positional coding is introduced to GlobalPointer's self-attention layer to improve its sensitivity to the sequence structure, and the loss function is optimized using the sharing matrix to reduce the number of parameters when increasing the entity types to complete the construction of the model. The experimental results show that the effect is significantly improved by using MBBLEGP hybrid model, and compared with the mainstream BERT-BiLSTM-CRF, the F1 value is improved by 2.21% and 1.21%, respectively, and the speed of convergence on F1 is improved, which verifies the effectiveness of the present model in the recognition of medical named entities.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing.

Keywords

Medical Named Entity Recognition, Natural language processing, location coding

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1 INTRODUCTION

Electronic Medical Record (EMR) is an electronic text record generated by hospitals and other related medical institutions in the process of diagnosis and treatment of patients, which is usually stored in an unstructured form. EMR contains a large number of patient-related diseases and medical treatment measures, etc. Mining these medical texts helps doctors to further diagnose and treat patients, and helps to improve the cure rate of patients[1].

Named Entity Recognition (NER) is a fundamental problem in Natural Language Processing (NLP), and the purpose of electronic medical record entity recognition is to automatically identify and label various medical entities from medical record records, such as treatment records, examination results, drugs, surgeries, nursing care, etc. This task is important for medical information management, clinical research, disease surveillance, and health care decision making [2].

Early NER tasks mainly recognized entities by based on lexical properties, grammatical architectures, lexicons, etc., and achieved some results [3]. Later with the development of deep learning, deep learning techniques can better capture the hidden features of text in the medical field, which improves the feature recognition effect of NER task. In recent years, with the wide application of deep neural networks, medical text entity recognition has achieved qualitative improvement in Long Short Term Memory (LSTM) and Conditional Random fields (CRF), but deep neural network models often fail to fully explore the feature information of the text. Due to the domain specificity of medical texts, entity recognition requires better understanding and processing of different types of entities, and medical entities have unique terminology and domain specificity. In order to better obtain text feature information, researchers have proposed the use of pre-trained models to realize the representation of word- and character-level vectors in text, among which the most representative model is the Bidirectional Encoder Representations from Transformer (BERT) [4]. It can be trained with corpus data from various domains, and the model is better at obtaining word vector representations from the sentence word and context levels [5].

In addition, most of the natural language processing downstream tasks using BERT pre-trained models achieve good results[6].BERT models have also had their development in the medical field.Facing the problem of the complexity of the text structure of the Chinese electronic medical records, the general pre-trained models in the general domain (e.g., BERT, etc.) are not able to represent the medical text very well[7].Furthermore, the self-attention mechanism is unable to capture the lexical elements in different positions .

The contribution of our work is set out below:

- The MBBLEGP (MC-BERT+BiLSTM+Efficient GlobalPointer) model is proposed, where the EHR data is input to MC-BERT for pre-training processing on both datasets, and the pre-trained word vectors are subjected to BiLSTM for extracting the long-distance textual semantic information, which is combined with the Efficient GlobalPointer for multi-label classification of feature vectors;
- Propose to introduce rotated position encoding in the attention mechanism to improve the ability of the attention mechanism to effectively utilize position information;
- Adopting a shared scoring matrix for the sequence labeling multiclassification scoring formula to reduce the number of parameters when increasing entity types.

2 REALATED RESEARCH

The NER task takes a sequence of characters as input and the output is a sequence of labels corresponding to the characters; it is a multicategorization sequence task. Nowadays, large language models and some multilayer neural networks have achieved excellent results in entity recognition, rule-based and dictionary-based NERs require professional linguists to develop templates and build specialized dictionaries, these methods have high precision but low recall [8].

Machine learning based approaches use supervised learning to transform NER tasks into classification or sequence labeling tasks, and this approach relies on feature construction. This mainly includes Hidden Markov Models (HMM), Conditional Random Field Models (CRF)[9], Support Vector Machines (SVM)[10]. This class of methods is more portable than dictionary and rule based methods.

Deep learning methods are used in NER tasks to solve automatic feature extraction on multilevel data. For example, neural network models based on Convolutional Neural Network (CNN), LSTM or BiLSTM, combined with machine learning models such as CRF, achieved the best results in NER tasks in different languages at that time.) and other neural network models, combined with machine learning models such as CRF, achieved the best results in NER tasks in different languages at that time[11][12].Zhang et al. proposed a lattice Long-Short-Term Memory network model for Chinese text categorization, which is able to add the meanings of the words themselves to the word-vector based model to reduce the effect of Chinese word-splitting errors[13]. Tang et al. proposed an attention-based model of CNN+LSTM+CRF, which was used to recognize Chinese medical clinical texts further improving the recognition effect[14].

The emergence of Transformer laid the foundation for entity recognition in the research of pre-training methods[15], and researchers proposed to use the representation learning method in deep learning for training.Early pre-training models for entity recognition mainly used symbolic discrete representations to transform words into One-Hot vectors for pre-training models. To address the problem of not being able to explain the multiple meanings of a word, Peters et al[16] proposed an ELMo model based on a two-layer BiLSTM structure to learn and train the relationship between context and word vectors.Radford et al[17] proposed a pre-training model based on the Transformer as an alternative to LSTM to obtain long-range information about a text.Devlin et al[18] proposed a BERT model based on bidirectional Transformer encoding for obtaining deep semantic information of this text, but it cannot fully utilize the explicit word and word order information. Current researchers obtain domain-specific pre-trained models by unsupervised learning on large-scale unlabeled data.

Zhang et al[19] proposed a Robustly Optimized Bert model pre-training Approach (RoBERTa-WWM) and applied it to the task of named entity recognition in electronic medical records with very significant results.Lee et al[20] in their study introduced a landmark model, BioBERT, which is the first pre-trained BERT model based on English medical text data.Fang et al[21] used the deep learning model RoBERTa-WWM-BiLSTM-CRF using RoBERTa-WWM, BiLSTM, and CRF models to efficiently solve the the problems of multiple meanings of words and incomplete word recognition in Chinese electronic medical record named entity recognition. A new conceptualization representation model was proposed by Ali's team, which pre-trained a MC-BERT model for the Chinese biomedical domain[22] and verified the effectiveness of the model on the newly proposed Chinese Biomedical Language Understanding Evaluation Benchmark (ChineseBLUE) dataset.

Pre-trained models such as RoBERTa-WWM, BioBERT, and MC-BERT have demonstrated superior performance in handling specific tasks in the medical domain. However, these models still have some shortcomings in processing long sequences of text and complex contextual information. Specifically, the current models are unable to capture long-distance dependency and sequential structural order features, are not sensitive enough to location information, and face the problem of parameter multiplication when adding entity types, which leads to the problems of low accuracy and difficulty in reducing computational complexity. In this paper, we improve the position encoding of the self-attention layer of Efficient GlobalPointer by introducing rotational position encoding to enhance the sensitivity of the model to the position information of the sequence structure, and optimize the loss function using the introduction of a shared scoring matrix to reduce the number of parameters when adding medical entities.

3 MEDICAL ENTITY RECOGNITION MODEL MBBLEGP

3.1 Problem definition

In the medical entity recognition task, the goal of medical entity recognition is to process textual data from a given medical record, which include electronic medical records, clinical notes, case reports, doctor-patient dialogues, etc. Various medical entities are recognized and labeled from the given sequence, including text, location, and entity type information. Specifically, the input is a

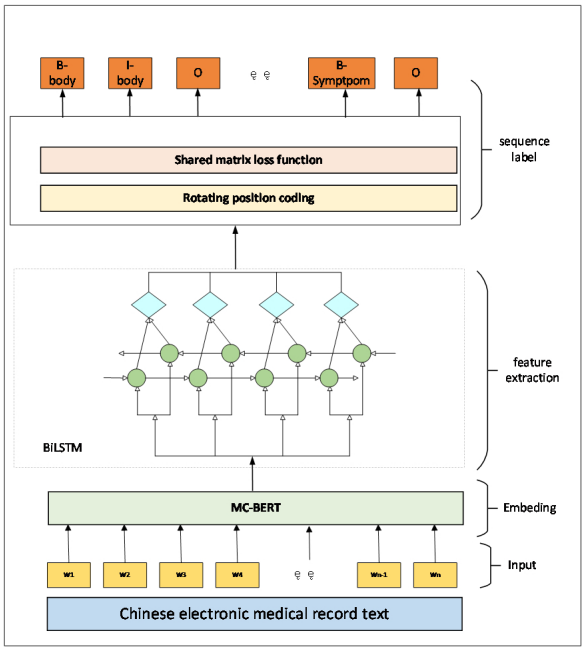

Figure 1: MBBLEGP model.

medical record sequence W=[w_1,w_2,w_3,...,w_{n-1},w_n] of text characters of length k (w_i denotes the corresponding word in the sentence, and n denotes the maximum length of the sentence), Each character w_i may be a Chinese character, a letter, or other text symbol. The medical record text may contain rich medical information, such as a description of the patient's condition, diagnosis results, and treatment plans.

After obtaining the extracted feature vectors, the Efficient GlobalPointer is used at the sequence annotation layer to classify the case entities using the positional information in combination with the representational information, and the entity types labeled at the sequence annotation layer are Y=[y_1,y_2,y_3,...,y_{n-1},y_n] (y_i for the corresponding diseases and diagnoses, surgeries, anatomies, symptoms, medications, and other medical entities).

3.2 MBBLEGP model

In this paper, an EHR named entity recognition model MBBLEGP is constructed by combining medical MC-BERT, BiLSTM and Efficient GlobalPointer model. as shown in Figure 1.

The basic process of MBBLEGP model Chinese electronic medical record entity recognition is as follows:

1)Input: firstly, the Chinese medical data text is extracted and the data format is converted to the format required by the model.

2)Embedding Representation: The processed data is input into the MC-BERT medical text recognition model, and the word-level and phrase-level semantic features in the text are captured by the Masked Medical Entity and Masked Medical Phrase strategies as inputs to the subsequent model.

3)Feature extraction: the output of MC-BERT is input into BiLSTM for sequence encoding, which selectively forgets or passes the electronic medical record text through the three structures of input gate, forgetting gate, and output gate, so as to capture the semantic dependencies of longer distances at the sentence level.

4) Sequence annotation: Finally, the improved Efficient GlobalPointer model predicts the nested entities by recognizing the "head" and "tail" features of the entities, so as to annotate the EHRs with BIO sequences. The labeling results are finally obtained.

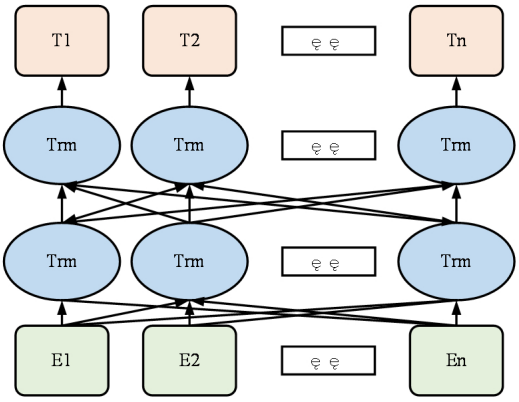

Figure 2: Structure of MC-BERT model.

3.3 Embedded Representation of MC-BERT

The MC-BERT has the same structure as the BERT structure, which consists of multilayer bi-directional Transformer coding, and the model framework is shown in Figure 2. En is the coded representation of the word. Trm is the Transformer structure, and Tn is the trained vector of the target word.The MC-BERT model performs unsupervised learning on a large amount of medical textual data, which better enable the model to learn the relationship between words, contextual semantic information and sentences[23].

Firstly, the text is converted into a vector representation at the word embedding layer using WordPiece disambiguation technique to ensure that each word can be represented efficiently, WordPiece disambiguation breaks down unregistered and rare words, into known sub-word units. Secondly, the word embeddings of the split words are processed through multiple layers of Transformer encoders, each layer of which contains feed-forward neural networks and multi-head self-attention mechanisms to compute the correlation between each word and generate weighted word representations, and feed-forward neural networks to nonlinearly transform the words that have been processed by the self-attention mechanisms. Breaking the last layer of Transformer encoder to extract the hidden states as the final feature representation.which are denoted as $V \in R^{k \times n \times t}$ (the size of t is generally 768, which denotes the size of the dimension of the self-attention header in BERT),These hidden state vectors contain contextual and semantic information about the input text.

3.4 BiLSTM feature extraction

Using the three gating structures of LSTM: input gate, forgetting gate and output gate, to solve the problem of gradient vanishing and gradient explosion generated by long text sequences during the

training process, the LSTM model also extracts valuable information from the neurons.The formulas of LSTM are given in in equation 1)-(6):

$$f_t = \sigma\left(w_f \odot h_{t-1} + u_f \odot x_t + b_f\right) \tag{1}$$

$$\mathrm{i}_t = \sigma\left(w_i \odot h_{t-1} + u_i \odot x_t + b_i\right) \tag{2}$$

$$\tilde{c_t} = \tanh\left(w_c \odot h_{t-1} + u_c \odot x_t + b_c\right) \tag{3}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c_t} \tag{4}$$

$$o_t = \sigma\left(w_o \odot h_{t-1} + u_o \odot x_t + b_o\right) \tag{5}$$

$$h_t = o_t \odot \tanh\left(c_t\right) \tag{6}$$

In the formula: f_t,i_t,h_t,o_t,c_t are the forgetting gate, the input gate, the hidden layer, the output gate and the unit state at the current moment; w_i,w_f,w_c,$w_o$are the weights of the previous hidden layer; u_i,u_f,u_c,$u_o$are the weights of the$x_t$current input; b_i,b_f,b_c,b_o are the relevant bias vectors, and $\tilde{c_t}$are the new candidate state vectors. $\sigma$is the sigmoid activation function and $\odot$is the dot product operation.

BiLSTM takes the representation vector $x_i$obtained from the embedded representation layer and processes the input sequence using one forward LSTM and one backward LSTM, computes the forward and backward hidden states $h^{(l)} = \{h_1^{(l)}, h_2^{(l)}, ..., h_n^{(l)}\}$and $h^{(r)} = \{h_1^{(r)}, h_2^{(r)}, ..., h_n^{(r)}\}$, and connects the two LSTMs to the same output layer, which can be formulated in equation 7).

$$\overleftrightarrow{h_t} = \tanh\left(W_h[h_t^{(l)} \oplus h_t^{(r)}] + b_n\right) \tag{7}$$

Each LSTM unit is capable of obtaining contextual information from the forward and backward terms, allowing BiLSTM to capture semantic dependencies of medical entities at longer distances in the vectors [24], and ultimately obtaining the feature vector matrix $V_B \in R^{h \times z}$ (h and z denote the dimensions of the vectors output from BiLSTM).

3.5 EGP Sequence Labeling with Improved Position Coding and Loss Function

Global Pointer Network Model[25] (Global Pointer, GP), EGP utilizes the global view for predicting the medical entity type when performing entity recognition, by designing two modules to recognize the start and end positions of the subject and object at the same time, thus achieving the discrepancy between the prediction and the training.The prediction of the EGP model is in parallel, which is more efficient compared to the CRF for dynamic planning.

EGP trains the head and tail of an entity as a whole when performing entity recognition, and EGP is equivalent to a simplified version of Multi-Head attention, which makes predictions through multiple heads, as shown in Figure 3. Among them, the entity type of "left hip" is the part, and the entity type of "left hip pain and discomfort" is the symptom.

The vectors obtained from the feature extraction layer are serialised as $[\mathrm{h}_1, h_2, \cdots, h_n]$. The vectors are input into the EGP for sequence annotation.

Where $q_{i,\alpha}$and $k_{i,\alpha}$ denote the start position characterisation information index and the end position characterisation information index, respectively, and $W_{q,\alpha}$ and $W_{k,\alpha}$are the corresponding

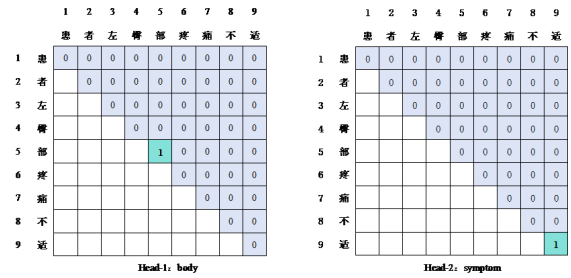

Figure 3: EGP Chinese Entity Prediction Map. The entity annotation statement in English is: The patient's left buttock is painful and uncomfortable

transformation matrices, respectively, Equation 8) and (9)below.

$$q_{i,\alpha} = W_{q,\alpha} h_i + b_{q,\alpha} \tag{8}$$

$$k_{i,\alpha} = W_{k,\alpha} h_i + b_{k,\alpha} \tag{9}$$

The scoring of consecutive characters from i to j entity type α in segment t is calculated by inner product of$q_{i,\alpha}$and$k_{i,\alpha}$. In order to fully consider the boundary information of the input character sequence, absolute position encoding is introduced in the entity scoring, and relative position encoding is realized by absolute position encoding on the ATTENTION mechanism, and the two-dimensional position encoding complex formula is expressed in terms of $f(\mathrm{q}, m)$ as in equation 10):

$$f\left(\mathrm{q}, m\right) = R_f\left(q, m\right) e^{i\theta_f(q,m)} = q e^{im\theta} \tag{10}$$

This complex form implies the rotatability of the vectors, which can be transformed from the complex in equation 11):

$$f\left(\mathrm{q}, m\right) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} q_0 \\ q_1 \end{pmatrix} \tag{11}$$

Since the inner product satisfies linear superposition, it can satisfy two-dimensional splicing in all even dimensional cases. $\mathfrak{R}_{j-i}$ is obtained by doing the attention mechanism processing on $\mathfrak{R}_{\mathrm{i}}$and $\mathfrak{R}_j$ to get the $s_\alpha(i, j)$-score in equation 12) based on the relative position encoding input sequence t, where $\mathfrak{R}_{\mathrm{i}}$ is an orthogonal matrix, and the orthogonal matrix does not change the stability of the model.

$$s_\alpha\left(i, j\right) = \left(\mathfrak{R}_{\mathrm{i}} q_{i,\alpha}\right)^T \left(\mathfrak{R}_j k_{j,\alpha}\right) = q_{i,\alpha}^T \mathfrak{R}_{j-i} k_{j,\alpha} \tag{12}$$

When every new entity type is added to EGP, two new transformation matrices $W_{q,\alpha}$ and $W_{k,\alpha}$ are added, and the number of parameters increases with the increase of entity types. In order to solve this problem, a shared matrix is proposed to solve the problem of increasing parameters, the same scoring matrix $(W_q h_i)^T (W_k h_j)$ is used for entity extraction of the same type of entity, and the parameters of entity extraction can be shared, and in the step of entity categorization, it is necessary to complete the "Feature Splicing + Dense Layer", and combine extraction and categorization to form a new The scoring function is shown in equation 13):

$$s_\alpha\left(i, j\right) = \left(W_q h_i\right)^T \left(W_k h_j\right) + w_\alpha^T \left[h_i; h_j\right] \tag{13}$$

When a new entity type is added, it is only necessary to increase the bias matrix of type α in the prediction of the entity type, that is,

Table 1: Parameters related to the data set.

Dataset	Train	Dev	Test
cEHRNER	914	44	41
Private dataset	6365	816	778

it is necessary to add new $w_\alpha \in R^{2D}$. The new addition of an entity type increases the number of parameters by 2D, so as to achieve the reduction of the number of parameters. Will $q_i = W_q h_i, k_i = W_q h_j$, $[q_i; k_i] = h_i$ to achieve a further reduction in the number of parameters, at this time $w_\alpha \in R^{4d}$, at this time, the number of parameters increased by 4d for each additional entity type, but due to$d \ll D$,so the number of parameters in equation 14) is smaller than equation 13), so the time complexity will be reduced.

$$s_\alpha(i, j) = q_i^T k_j + w_\alpha^T \left[q_i; k_i; q_j; k_j\right] \tag{14}$$

4 EXPERIMENTAL RESULTS AND ANALYSIS

4.1 Data sets

For a fairer and more comprehensive comparison, the experiments in this paper use the publicly available dataset cEHRNER and the clinical dataset from a tertiary hospital (Internal Medicine Data) as shown in Table 1. cEHRNER is the ChineseBLUE (Chinese Biomedical Language Understanding Evaluation Benchmark) of the new Chinese language released by Ali's team in the MC-BERT paper. EMR dataset, including entities such as diseases, drugs, and syndromes, where the data are selected from the tagged dataset in Chinese electronic health records. Clinical dataset The clinical dataset from the hospital is processed into a desensitized private dataset for the entity identification task, in order to ensure data privacy and security, as well as to ensure the quality and validity of the data, the text of the collected cases is de-noised such as invalid records, blank fields, and duplicates, and at the same time, the personally identifiable information is anonymized or removed, and the common personally identifiable information includes name, ID number, contact information, address, etc.

4.2 Experimental setup and assessment indicators

Experiments were conducted using MC-BERT model to pre-train the dataset, based on three different pre-training models BERT[26](2021), RoBERTa, BioBERT and MC-BERT for comparison,with BERT-BiLSTM-CRF, RoBERTa-wwm-BiLSTM-CRF[27], BioBERT-BiGRU-CRF[28], ClinicalBERT-CNN-BiLSTM-CRF[29] and other models to do comparison experiments, ten repetitions in the same experimental environment to verify the effect of the present model, the experimental model parameters are set as shown in Table 2. In order to avoid the overfitting phenomenon, all the experimental models use the AdamW optimizer. The AdamW optimiser corrects the weight decay problem in the Adam optimiser, for the selection of hyperparameters such as learning rate, weight_deca, batch size, etc., we carried out several experiments, and finally chose the results in Table 2, which will help to prevent overfitting.

Table 2: Model parameter settings.

Parameter	Value
BiLSTM vector dimension	512
Batch_size	64
Learning rate	1e-5
Sentence max length	256
Dropout	0.4
Epochs	30
Optimizer	AdamW
AdamW learning rate	2e-4
AdamW weight_decay	1e-5
AdamW batch size	64

Table 3: NER results for different pre-trained models on cEHRNER.

BERT models	p	R	F1
BERT-EGP	89.05	89.76	89.37
RoBERTa-EGP	90.17	90.68	90.3
BioBERT-EGP	90.32	91.24	90.54
MC_BERT-EGP	91.14	91.85	91.46

Table 4: NER results for different pre-trained models on Internal Medicine Dataset.

BERT models	P	R	F1
BERT-EGP	90.72	90.75	90.41
RoBERTa-EGP	91.19	91.96	91.5
BioBERT-EGP	91.42	92.13	91.63
MC_BERT-EGP	93.54	92.62	93.11

4.3 Experimental results and analysis

In order to compare the effectiveness of different pre-trained BERT models for entity recognition, this paper uses different versions of BERT pre-trained models and MC-BERT models for performance comparison. The experimental results are shown in Tables 3 and 4

Bold in all tables is the maximum value of the column value. From the experimental results in Table 3 and Table 4, there is a difference in the performance of different pre-training models in the Chinese EHR NER.EGP is used to decode the feature vectors of the pre-training outputs. The F1 of the MC-BERT+EGP model reaches the highest 91.46 and 93.11, which indicates that the MC-BERT+EGP model has a higher accuracy and recall in the Chinese pre-training model has higher accuracy and recall, which can better improve the entity recognition of Chinese EHRs.

In order to show the effectiveness of this model compared to other Chinese EHR entity recognition models, this model is compared with NER models in different literatures by conducting ten repetitions and taking the optimal results under the same experimental experimental setup, and the final experimental results are shown in Tables 5 and 6

Table 5: Results of different NER models on cEHRNER.

Models	P	R	F1
BiLSTM-CRF	84.82	85.21	85.07
BERT -CRF	88.75	89.52	89.14
BERT-BiLSTM-CRF	89.39	89.95	89.72
RoBERTa- BiLSTM-CRF	90.18	91.16	90.67
RoBERTa-wwm-BiLSTM-CRF	91.73	90.21	90.94
BioBERT-BiGRU-CRF	91.43	91.53	91.48
ClinicalBERT-CNN-BiLSTM-CRF	91.65	91.34	91.5
MC_BERT- BiLSTM-EGP	92.12	92.33	92.27

Table 6: Results of different NER models on Internal Medicine Dataset.

Models	P	R	F1
BiLSTM-CRF	84.77	87.67	86.54
BERT -CRF	89.26	89.73	89.49
BERT-BiLSTM-CRF	91.72	91.98	91.84
RoBERTa- BiLSTM-CRF	92.36	92.59	92.43
RoBERTa-wwm-BiLSTM-CRF	92.14	92.87	92.51
BioBERT-BiGRU-CRF	92.7	92.44	92.65
ClinicalBERT-CNN-BiLSTM-CRF	92.81	92.6	92.77
MC_BERT- BiLSTM-EGP	93.87	92.35	93.42

From the experimental results of the two datasets in Tables 5 and 6, compared with the base BiLSTM-CRF model, the BERT -CRF model can obtain better feature representations by pre-training the EHR data, and the values of the evaluation metrics P, R, and F1 on the cEHRNER dataset and the Internal Medicine Dataset dataset are respectively improved by 3.93%, 4.31%, 4.07% and 4.49%, 2.06%, 3.36%. Compared to the BERT model BERT-WWM model obtains the word vector representation of the EHR by using the all-word masking strategy, and from the results of the experiments comparing the BERT-WWM- BiLSTM-CRF model and the BERT-BiLSTM-CRF model, each evaluation metric has been improved, and the F1 has been improved by 0.95% and 0.56%, respectively. By comparing the experimental results using RoBERTa-wwm-BiLSTM-CRF model and BERT-WWM- BiLSTM-CRF model, the experimental results show that the F1 values are improved by 1.22% and 0.67% respectively, which indicates that the RoBERTa-wwm model has a better result than the BERT model in entity recognition, using dynamic masking instead of static masking strategy and generating a separate mask for each input and learning different linguistic features with different masking strategies as the training progresses.

In the medical MC_BERT model, the MC_BERT- BiLSTM-CRF model has better recognition effect compared with the pre-trained models in recent years, the MC_BERT model is trained by using the proprietary corpus of the medical domain, and the medical MC_BERT model has significant improvement in mining medical text entities compared with other advanced BERT models, the RoBERTa-wwm model has better vector representation effect compared with the BERT-BiLSTM-CRF model. RoBERTa-wwm model has a better vector representation by training with a corpus such as Chinese Wikipedia, and MC_BERT- BiLSTM-CRF model improves the P, R, and F1 values by 2.42%, 2.2%, and 2.21%, and 1.22%, 1.14%, and 1.21%, respectively, compared to the BERT-BiLSTM-CRF model.

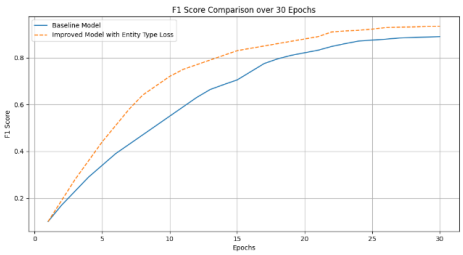

Figure 4: Effect of shared parameter matrix loss function on convergence rate.

The merit of the loss function directly determines how well the model fits the parameters, and the convergence speed of the training process is accelerated by adding a shared parameter matrix to the loss function. Figure 4 demonstrates the effect of the shared parameter matrix loss function on the convergence speed. The results of this experiment are based on the experimental setup in Table 2, and the results of the comparative experiments are obtained by improving the loss function of this model only. Twenty experiments were conducted on this dataset separately and the experimental F1 values were recorded for each round. From the experimental results, it can be seen that using the loss function with shared parameter matrix can significantly improve the performance of the raw model while accelerating the convergence speed of the model.

5 CONCLUSION

In this study, this paper proposes a new hybrid model based on the combination of MC_BERT, BiLSTM and EGP to extract entities from Chinese EMR. The method firstly uses the Chinese pre-training model MC_BERT as the word vector representation of Chinese EMR, utilizes the three gating mechanisms of BiLSTM model to obtain semantic dependencies over longer distances, so as to obtain more accurate semantic features of EHRs, and then combines with the EGP model, introduces rotational positional encoding to the positional encoding in the ATTENTION mechanism to effectively utilize the positional information The model effectively solves the problem of nested complexity of Chinese EHR entities by using shared parameter matrix for sequence labeling multiclassification scoring formula to solve the problem of multiplying the number of parameters when increasing the entity types. The model effectively solves the problem of nested entities in Chinese electronic medical records. Better recognition results are achieved on both the public dataset cEHRNER and the private dataset Internal Medicine Dataset, which provides certain reference value for entity recognition in Chinese electronic medical records.

References

[1] Nasar Z, Jaffry S W, Malik M K. Named entity recognition and relation extraction: State-of-the-art[J]. ACM Computing Surveys (CSUR), 2021, 54(1): 1-39.

[2] Wu S, Song X, Feng Z. MECT: Multi-metadata embedding based cross-transformer for Chinese named entity recognition[J]. arXiv preprint arXiv:2107.05418, 2021.
[3] An Y, Xia X, Chen X, *et al.* Chinese clinical named entity recognition via multi-head self-attention based BiLSTM-CRF[J]. Artificial Intelligence in Medicine, 2022, 127: 102282.
[4] Vaswani A, Shazeer N, Parmar N, *et al.* Attention is all you need[J]. Advances in neural information processing systems, 2017, 30.
[5] Chen P, Zhang M, Yu X, *et al.* Named entity recognition of Chinese electronic medical records based on a hybrid neural network and medical MC-BERT[J]. BMC Medical Informatics and Decision Making, 2022, 22(1): 315.
[6] Zhou S, Liu J, Zhong X, *et al.* Named entity recognition using BERT with whole world masking in cybersecurity domain[C]//2021 IEEE 6th International Conference on Big Data Analytics (ICBDA). IEEE, 2021, 316-320.
[7] Wen C, Chen T, Jia X, *et al.* Medical named entity recognition from un-labelled medical records based on pre-trained language models and domain dictionary[J]. Data Intelligence, 2021, 3(3): 402-417.
[8] Song B, Li F, Liu Y, *et al.* Deep learning methods for biomedical named entity recognition: a survey and qualitative comparison[J]. Briefings in Bioinformatics, 2021, 22(6): bbab282.
[9] Eddy S R. Hidden markov models and large-scale genome analysis[J]. Transactions of the American Crystallographic Association, 1997.
[10] Cortes C, Vapnik V. Support-vector networks[J]. Machine learning, 1995, 20: 273-297.
[11] Ma P, Jiang B, Lu Z, *et al.* Cybersecurity named entity recognition using bidirectional long short-term memory with conditional random fields[J]. Tsinghua Science and Technology, 2020, 26(3): 259-265.
[12] Tang B, Wang X, Yan J, *et al.* Entity recognition in Chinese clinical text using attention-based CNN-LSTM-CRF[J]. BMC medical informatics and decision making, 2019, 19: 89-97.
[13] Zhang Y, Yang J. Chinese NER using lattice LSTM[J]. arXiv preprint arXiv:1805.02023, 2018.
[14] Tang B, Wang X, Yan J, *et al.* Entity recognition in Chinese clinical text using attention-based CNN-LSTM-CRF[J]. BMC medical informatics and decision making, 2019, 19: 89-97.
[15] Vaswani A, Shazeer N, Parmar N, *et al.* Attention is all you need[J]. Advances in neural information processing systems, 2017, 30.
[16] Peters M E, Neumann M, Iyyer M. Deep contextualized word representations [EB/OL][J]. arXiv preprint arXiv:1802.05365, 2018.
[17] Radford A, Narasimhan K, Salimans T, *et al.* Improving language understanding by generative pre-training[J]. 2018.
[18] Devlin J, Chang M W, Lee K, *et al.* Bert: Pre-training of deep bidirectional transformers for language understanding[J]. arXiv preprint arXiv:1810.04805, 2018.
[19] Zhang X, Zhang Y, Zhang Q, *et al.* Extracting comprehensive clinical information for breast cancer using deep learning methods[J]. International journal of medical informatics, 2019, 132: 103985.
[20] Lee J, Yoon W, Kim S, *et al.* BioBERT: a pre-trained biomedical language representation model for biomedical text mining[J]. Bioinformatics, 2020, 36(4): 1234-1240.
[21] Fangcong Z, Qiuli Q, Yong J, *et al.* Named Entity Recognition for Chinese EMR with RoBERTa-WWM-BiLSTM-CRF[J]. Data Analysis and Knowledge Discovery, 2022, 6(2/3): 251-262.
[22] Yuan Z, Zhao Z, Sun H, *et al.* CODER: Knowledge-infused cross-lingual medical term embedding for term normalization[J]. Journal of biomedical informatics, 2022, 126: 103983.
[23] Hou Z, Salazar J, Polovets G. Meta-learning the difference: preparing large language models for efficient adaptation[J]. Transactions of the Association for Computational Linguistics, 2022, 10: 1249-1265.
[24] Jin C, Shi Z, Li W, *et al.* Bidirectional lstm-crf attention-based model for chinese word segmentation[J]. arXiv preprint arXiv:2105.09681, 2021.
[25] Su J, Murtadha A, Pan S, *et al.* Global pointer: Novel efficient span-based approach for named entity recognition[J]. arXiv preprint arXiv:2208.03054, 2022.
[26] Ye Y N, Zheng L F, Huang M Y, *et al.* The algorithms for word segmentation and named entity recognition of chinese medical records[C]//Advances in Artificial Intelligence and Security: 7th International Conference, ICAIS 2021, Dublin, Ireland, July 19-23, 2021, Proceedings, Part I 7. Springer International Publishing, 2021: 397-405.
[27] Liu Z, Lin W, Shi Y, *et al.* A robustly optimized BERT pre-training approach with post-training[C]//China National Conference on Chinese Computational Linguistics. Cham: Springer International Publishing, 2021: 471-484.
[28] Sun C, Yang Z, Wang L, *et al.* Biomedical named entity recognition using BERT in the machine reading comprehension framework[J]. Journal of Biomedical Informatics, 2021, 118: 103799.
[29] Li Y, Wehbe R M, Ahmad F S, *et al.* A comparative study of pretrained language models for long clinical text[J]. Journal of the American Medical Informatics Association, 2023, 30(2): 340-347.